

AI Training

Diego Pacheco

• 2026

▪ AI, Agents and People

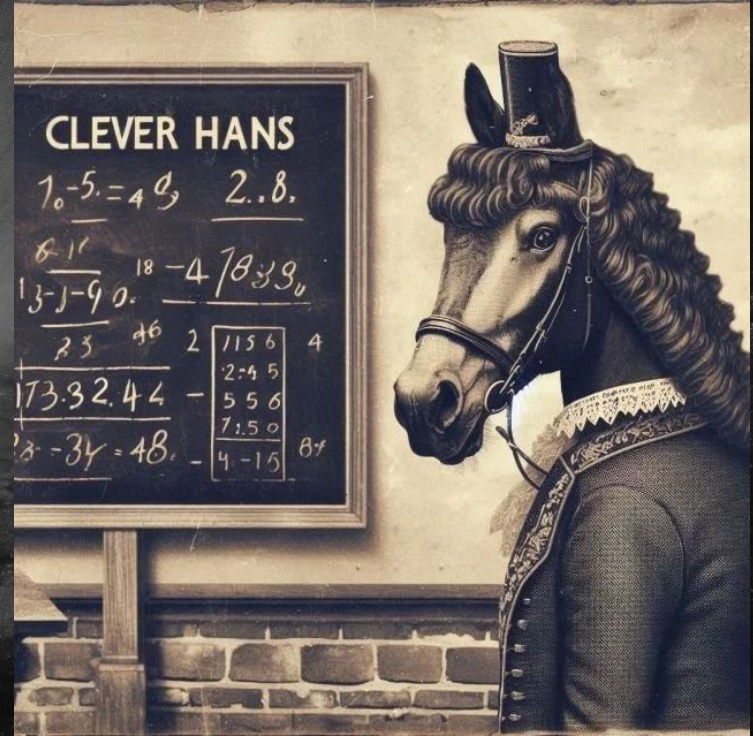
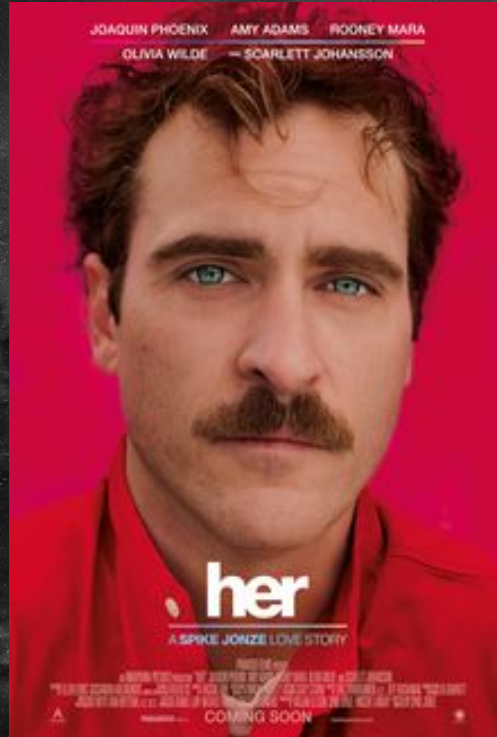
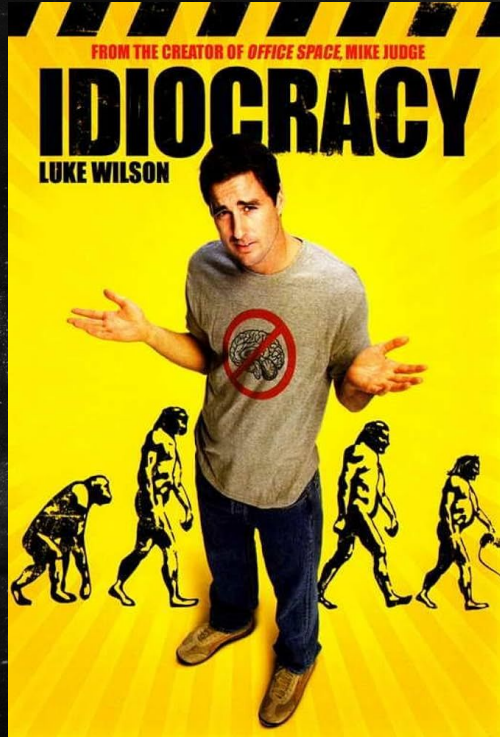
Part 1

AI

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Pre-Requirements



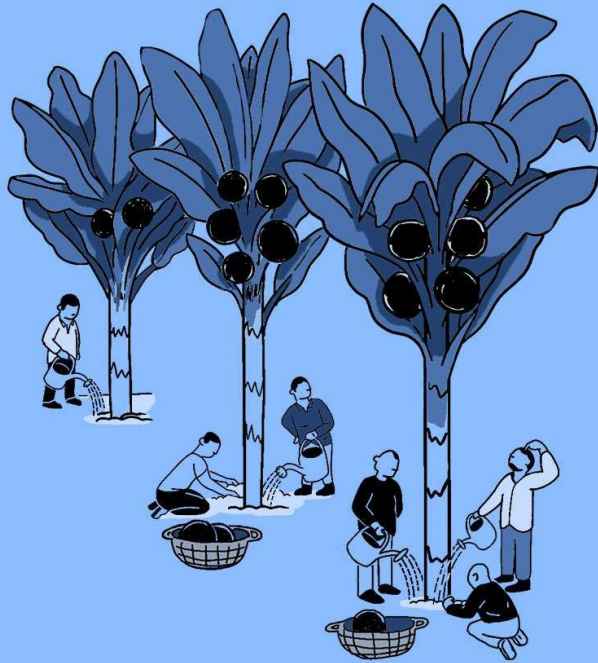
Molts, Lobsters and other animals...



<https://www.moltbook.com/>

- Scams, Fakes, Humans...
- Security Nightmare...
- Molt Church, Agent Encryption, Molt Lang, Molt selling humans...
- What it means manage people and agents?
- Are molts new customers?
- Molt Experience (MOX)?
- Molt Laws?
- Acquisitions (OpenAI, Meta)

Hitting limits...



Law of Diminishing Marginal Returns

['lɒ əv də-'mi-nish-ɪŋ 'mɑːrj-nəl ri-'təʊns]

A theory in economics that predicts that after some optimal level of capacity is reached, adding an additional factor of production will actually result in smaller increases in output.

- More Hardware
- Maybe nuclear?
- Need a break through
- Cost Race to the bottom
- Tokens explored due "reasoning" models...

Hitting limits...

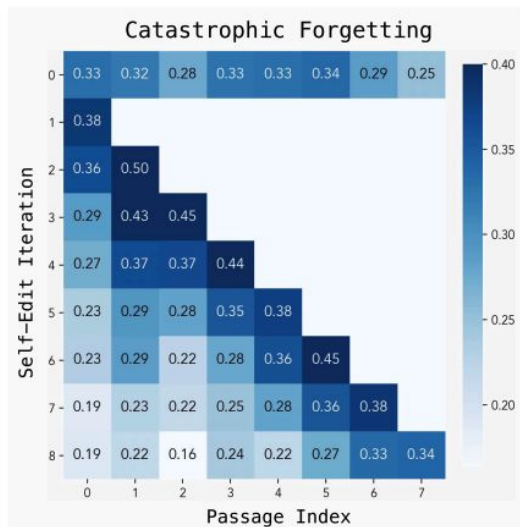
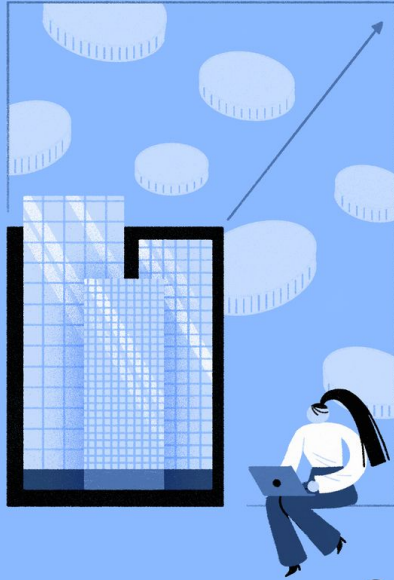


Figure 6: **Catastrophic forgetting from continual self-edits.** We sequentially update the model on new passages and track degradation on prior tasks. Entry-wise standard errors are reported in §B.6.

- I know is called Machine Learning
- But only during "Training phase"
- LLM Don't Learn
- They memory is a markdown file
- You repeat over and over...
- Engineering and guardrails to rescue...

Hitting limits...



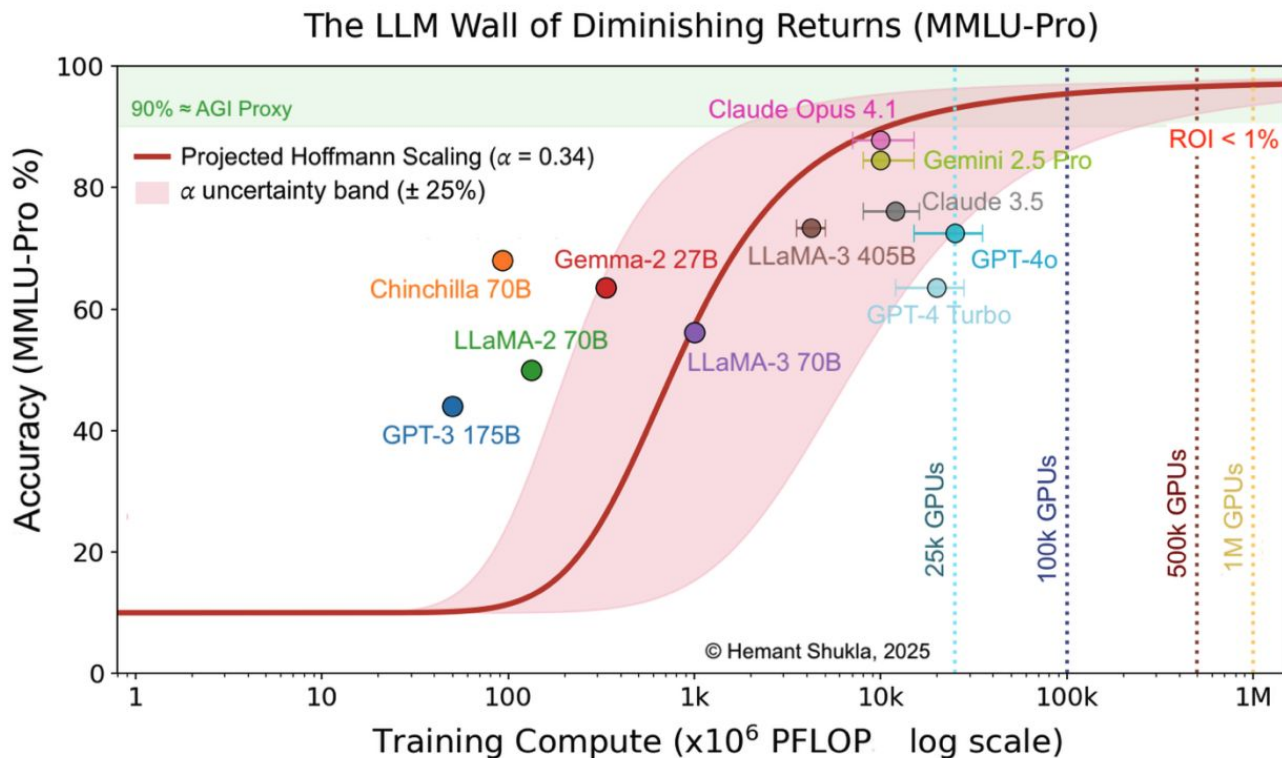
Scalability

[,skā-lə-'bi-lə-tē]

The capacity of a company to adapt to and profit from an increased demand for its products or services.

- Still about training
- We don't have another internet
- Self Learning decay...
- LLM models are marginally better
- Engineering to rescue
- OSS to rescue

Hitting limits...



How much % automation?



- AI Barista
- Jarvis
- Artly: Robots
Serving Coffee in
San Francisco
- Dude behind the
scenes
- vs whole ai coffee
shop...

Cost of POC



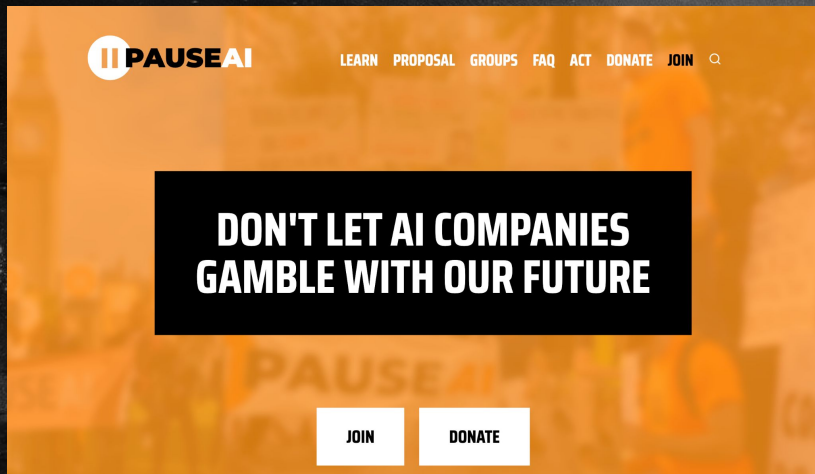
Backpressure or pay more..



You've hit your limit · resets 8pm
(America/Los_Angeles)



Anti-AI & OSS



<https://pauseai.info/>



open source

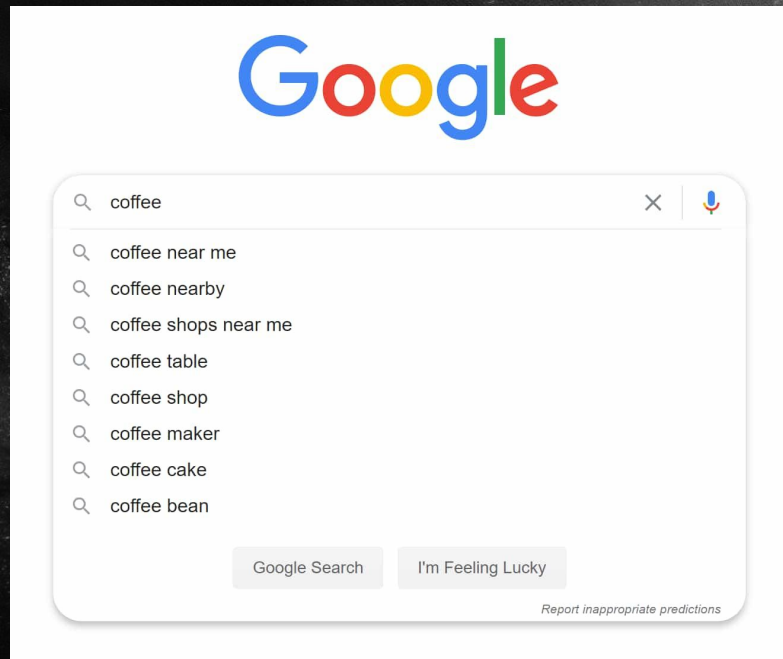
AI Ban or Strict Policies:

- Zig
- Curl
- Ghostty
- Gentoo Linux / FreeBSD
- Tldraw
- Qemu
- LLVM

What is an LLM?

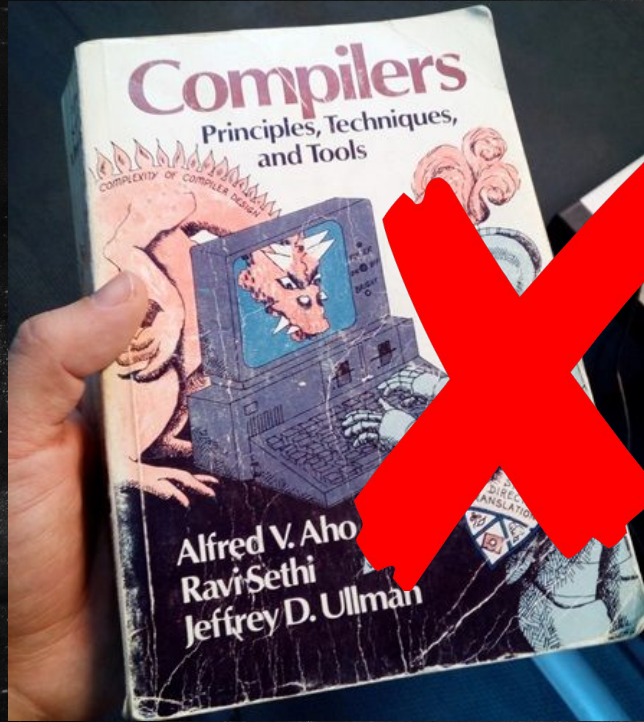


What is an LLM?



Next Token Prediction Machine == Auto-complete

What is an LLM?



- LLM are not compilers

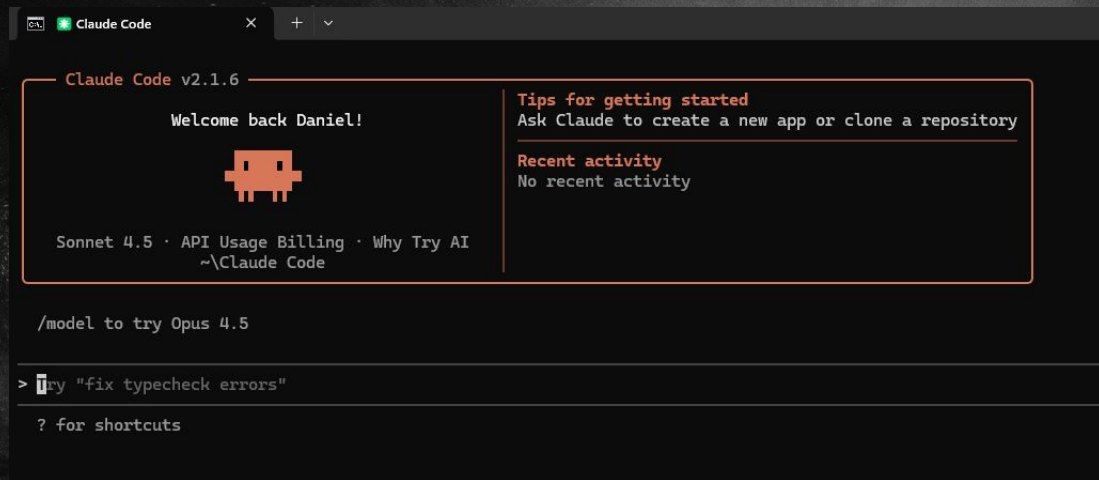
Part 2

Agents

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▪ AI, Agents and People

Claude Code



<= Code Agent

<= Multi-Agents

Claude Code is the new IDE.

Skill Diff



Side Effects: Less patience



Side Effects: More operational, more execution



- Claude code Dopamine.
- It's faster to do than to think hard.
- Fight our instincts.
- Do non-obvious things...

Does anyone know Waterfall?



- Waterfall
- Never died
- It's back
- SDD
- Specs
- SAFE as Agents :(

Does anyone know Waterfall?



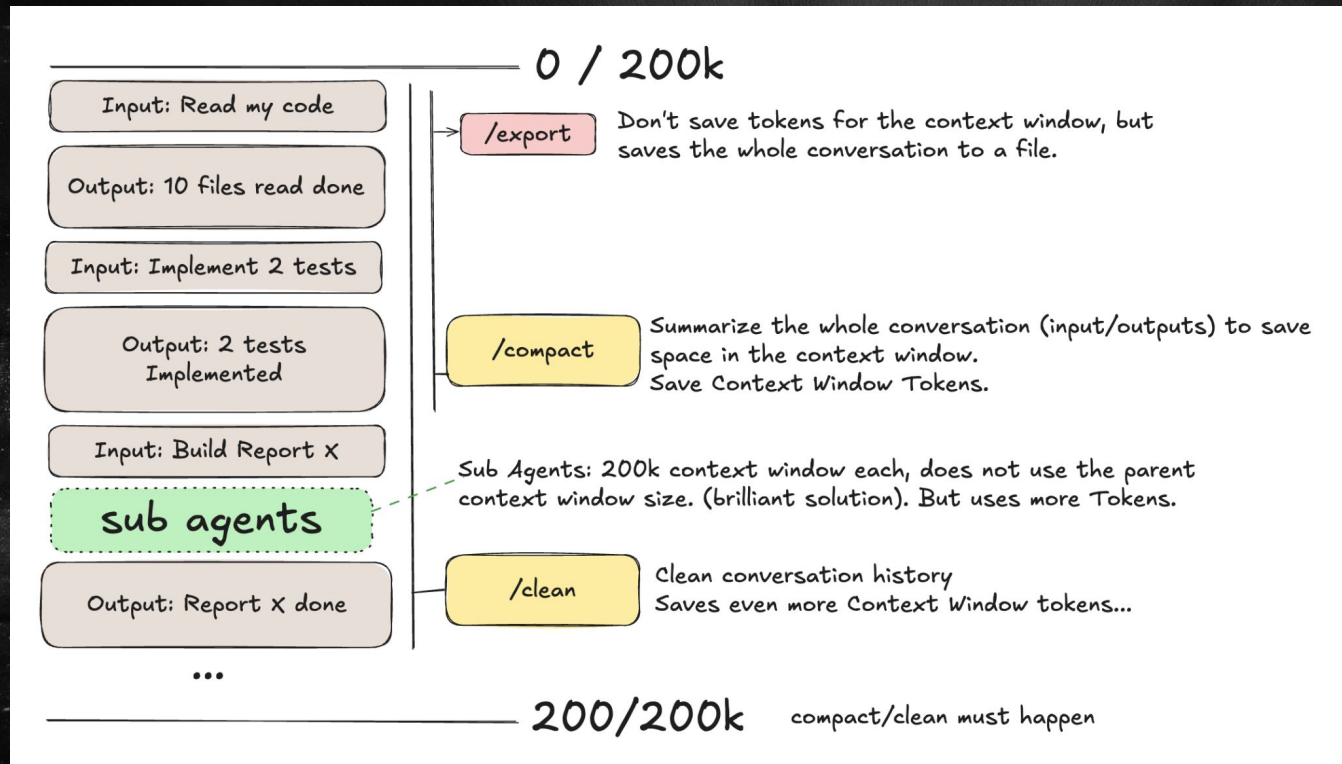
```
Need to install the following packages:  
bmad-method@4.43.0  
Ok to proceed? (y) y
```

BMAD-METHOD

```
🔥 Universal AI Agent Framework for Any Domain  
★ Installer v4.43.0
```

```
? Enter the full path to your project directory where BMad should be installed:
```


Does anyone know Waterfall? Context for all !!!



- Context rot
==
game over
- Model gets
dumber.
- Be aware
- Fight it...

Does anyone know Waterfall?

Spec Driven Development (SDD) -> Sucks!

More text $\neq$ better results.

Large specs create:

- Context bloat
- Attention dilution
- Instruction collapse
- Maintenance overhead
- Stale documentation

Real Issues:

- Agents "choking on sheer volume"
- Instructions ignored or misread
- Older sections conflicting with new ones
- Slow, inconsistent execution

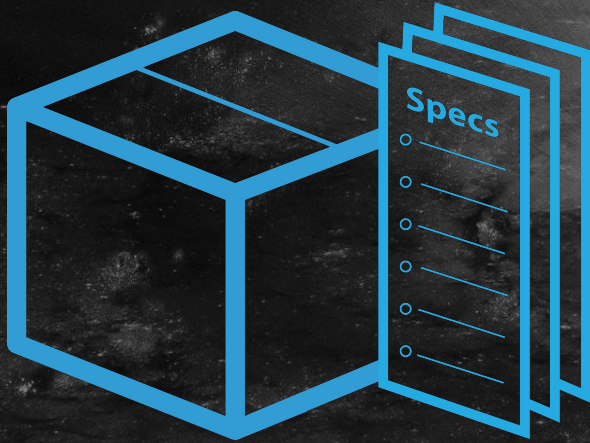
Are we back to Waterfall?

Key evidence:

- Diminishing returns after 2–3 examples
- Over-prompting causes performance drop
- Long specs confuse models instead of helping
- Short prompts outperform in simple tasks

- Docs/UML-heavy processes
- Model-Driven Development
- Big upfront spec systems
- Problems: rigidity, drift, overhead.
- Stale documents
- More Text $\neq$ More Clarity

Bad Trade

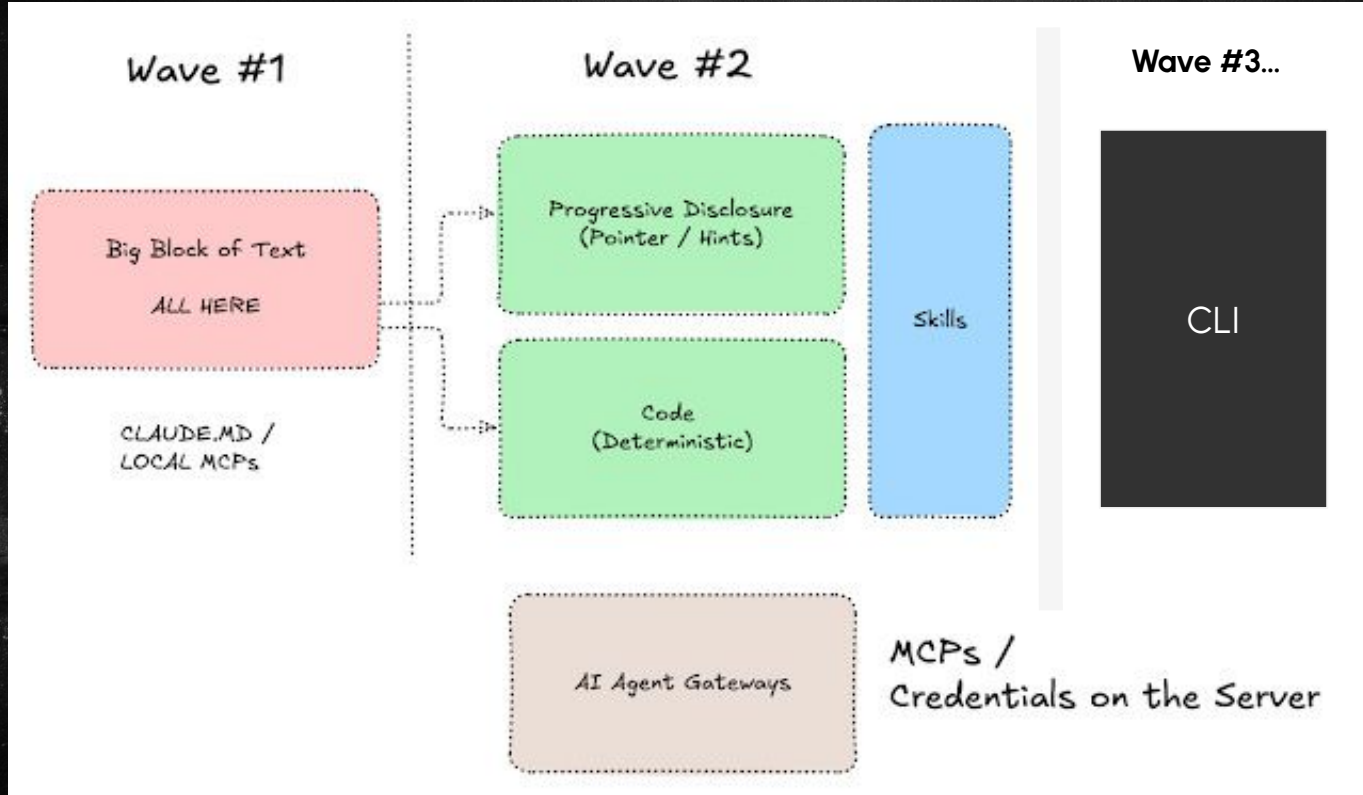


not a context-free grammar



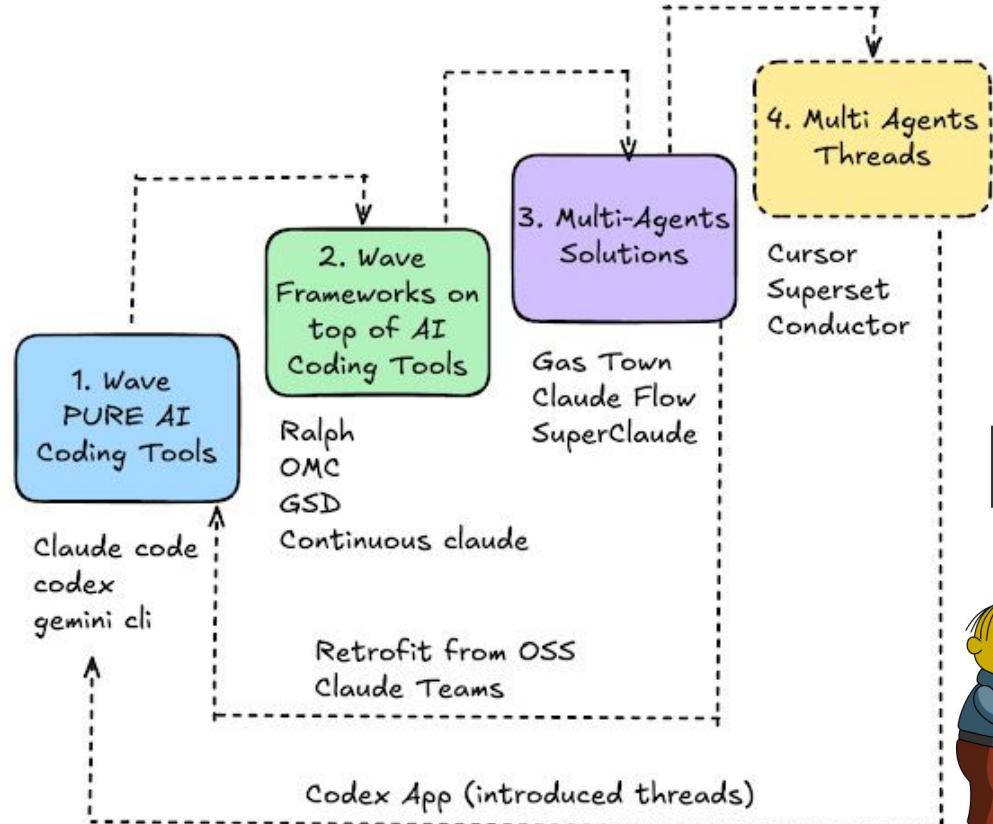
Deterministic

Context Window Usage Evolution



Claude Code Evolution

CLAUDE
CODE



Claude Code Evolution



<https://diegopachecotech.substack.com/p/ai-coding-agents-evolution>

Does anyone know Waterfall?

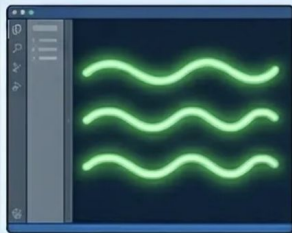


Figure 1



Figure 2



Figure 3



Figure 4



Figure 5



Figure 6



Figure 7



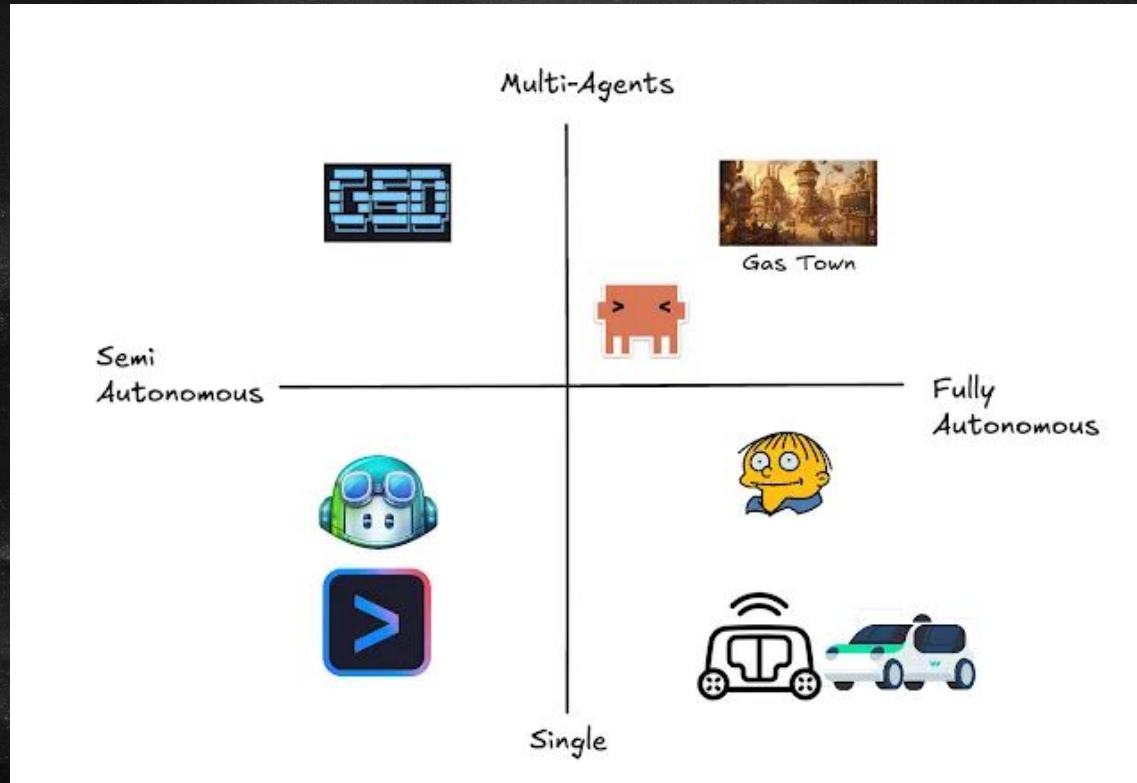
Figure 8

Figure 2: The 8 Stages of Dev Evolution To AI

Gas Town: Kubernetes of multi-agents



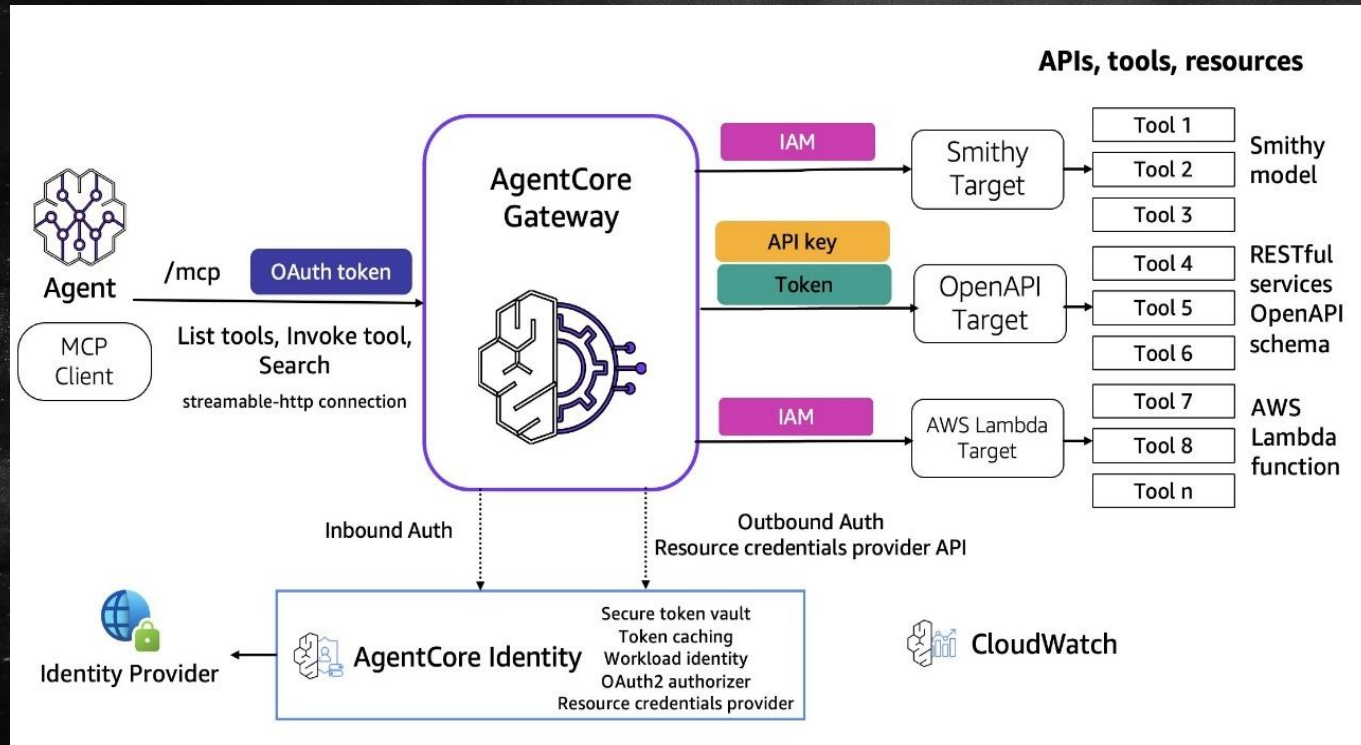
Multi-Agents



Multi-Agents: Consequences

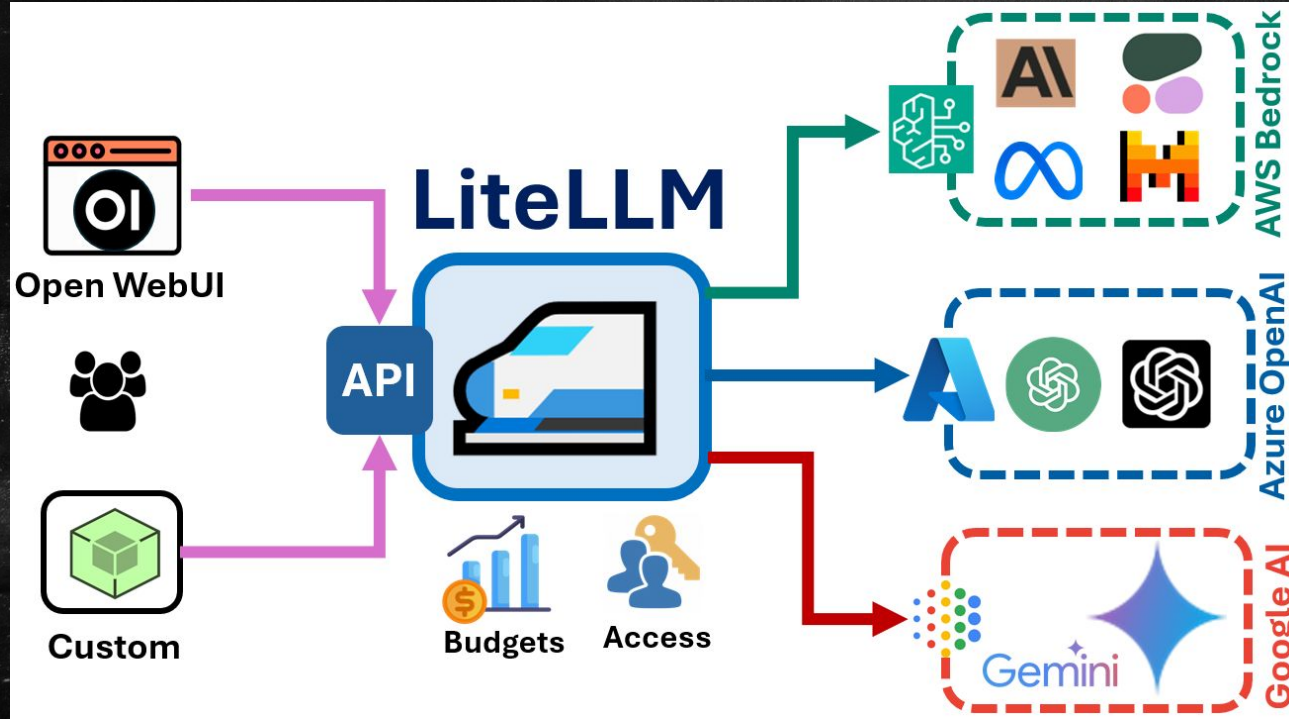


Multi-Agents: Productivity or Production?



- Auditability
- Observability
- Cost Control
- AIM
- Tagging
- Security

Multi-Agents: Productivity or Production?



- Observability
- Caching
- Fallover
- Fallback
- Cost savings
- Routing...

Part 3

People

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Side Effects: Demo vs Production



2014



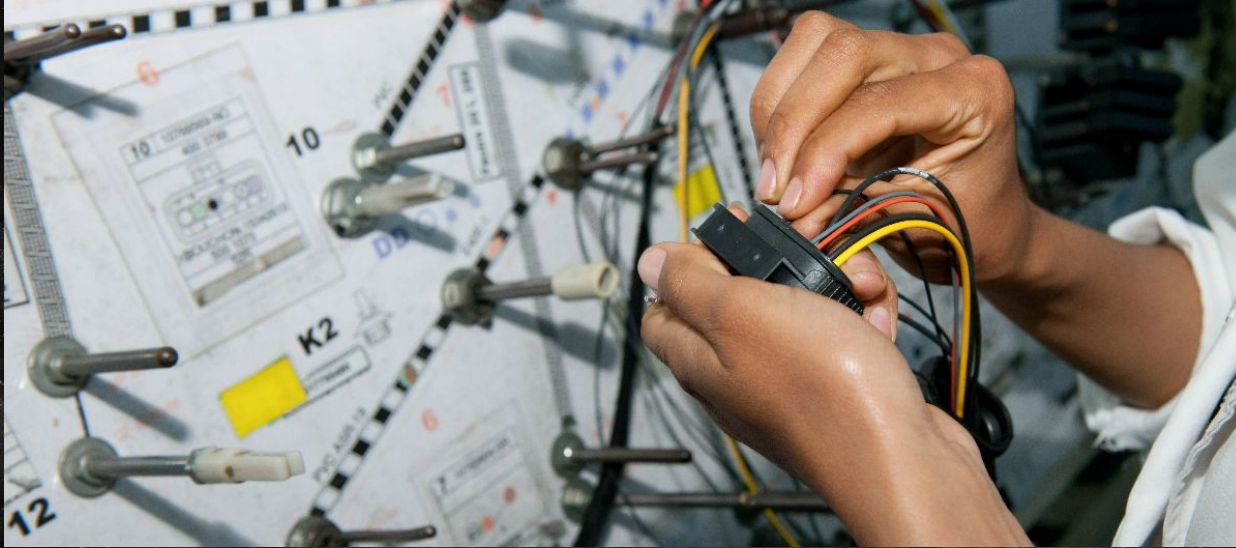
2025

Skills that Matter: Critical Thinking



- (I) should I be doing this?
- Any other options?
- Is this right?
- Does the code looks good?
- Tests are good?

Skills that Matter: Attention to Detail



- Look the code
- Look the code
- Look the code
- Run the code
- Run the tests
- Think...
- Think...
- Think...

Skills that Matter: Creativity



- What can I do now that before was expensive or impossible?
- What nobody is doing?
- How to break the standards?
- Scenario thinking (like UX).
- External Inspiration Sources

Skills that Matter: Know how it works... and Concepts



- DO I understand?
- Can I explain with clarity?
- Do I know the difference?
- How do I test it?

Safety for People and Agents: Guardrails

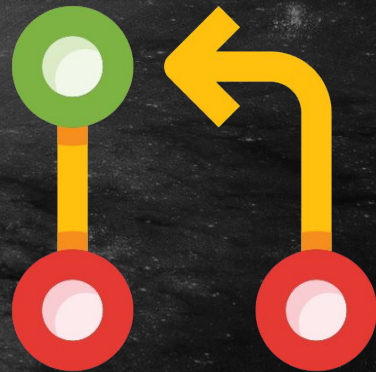


- AI Shift Left
- Testing
- More Automation
- More Test Diversity
- Policy as Code

Safety for People and Agents: Backpressure

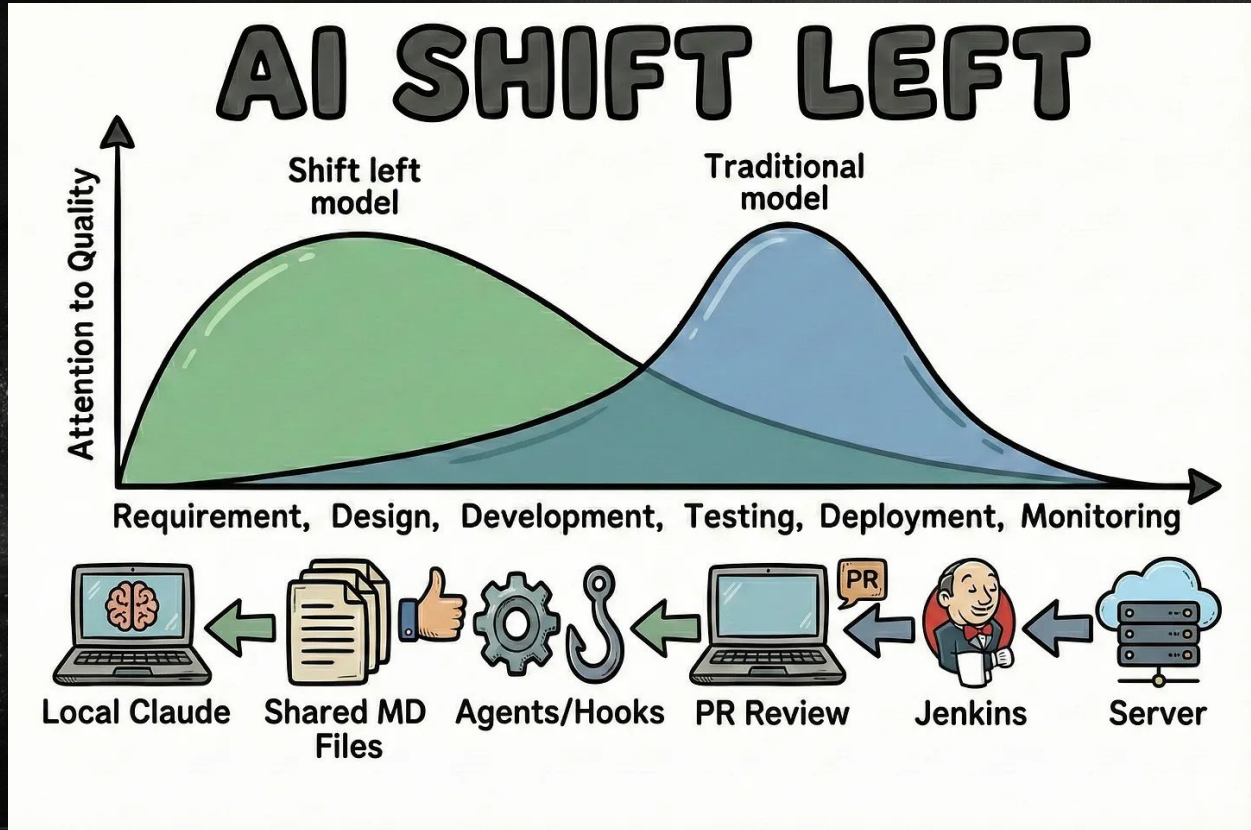


Code Review is backpressure



Prompt Requests?

Safety for People and Agents: AI Shift Left



- All in the Dev Machine.
- How can we do that with requirements?
- What about UX?

Let's not forget: Interactive => Chat



The rise of The Architect!



- Understand how it works
- Concepts
- Decisions
- Contracts
- Design
- Security
- Overall Architecture
- Code Review off cycle

Time to Work Hard !!!



Time to Work Hard !!!



All Exercises with claude code only!

1. Create a CLI application using claude to lint your emails. (max 3 LoC)
2. Workflow: Ask claude to not code anything, ask him to write an design-doc.md when you done, define, architecture, code style, guidelines, describe screens, them ask hilton break in tasks on todo.txt and finally to build (Build a retrospective application).
3. Using 1 terminal, 1 claude code session make claude spin 3 agents in parallel. Build a frontend/backend forum application. Can you spin more than 3 now?
4. Build a skill to create changelogs on the application using code and git history (run on the forum application).
5. Build a multi-agent system to perform code review with AI shift-left.
6. Build a multi-agent system to create an onboarding game(quiz) about a given service, should have frontend and backend.

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